

Hexgame with Reinforcement Learning

By Group F

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Agent's strategy: Alphazero

AlphaZero MCTS in Hex pairs a neural network with Monte Carlo Tree Search. The network reads the board (from the side-to-move's view — handy in Hex since `recode_blue_as_red()` lets Blue reuse a Red-trained net) and outputs a policy (move priors) and a value (who's winning). MCTS then builds a search tree, descending via the PUCT rule that balances the average value of a move against its network prior and visit count; at each leaf it uses the network's value estimate rather than a random playout, expands the children, and backpropagates the value with a sign flip at each level (zero-sum). After a fixed number of simulations, it plays the most-visited move. The network is trained by self-play, where the search's visit counts become policy targets and game outcomes become value targets, feeding stronger searches in a self-improving loop — a great fit for Hex since it never draws and is symmetric across colors.

The Model and code

The directory contains only the model for the board size 5, 7 and 9 due to the maximum upload in Moodle. Please download the completed models at https://github.com/philippchn/RL-Hexgame/raw/refs/heads/cahya/group_f.tgz